Introductory Mathematical Economics

Bound

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Reading: Will be uploaded on Piazza

Outline

- Definition of upper bound and lower bound
- Definition of supremum and infimum
- Alternative definition (ϵ- definition) of supremum and infimum
- Equivalence of two definitions
- Algebra of supremum/infimum

Upper Bound

Let S be a non-empty set of real numbers.

Suppose there is a real number B such that $B \geq x \, \forall x \in S$. Then B is an **upper bound** for S. S is said to be **bounded above** by B.

Observation 1: There can be many upper bounds of a set.
Example: $S = (0, 4]$. Both 5 and 4 are upper bound of S.

Observation 2: B need not be a member of S.
Example: $S = (0, 4]$. Upper bound 5 is not in S.

A set which has no upper bound is said to be **unbounded above**.
Example: Set of all positive integers is unbounded above.

Supremum

Let S be a non-empty set of real numbers.

A real number B is called a **least upper bound** (or **supremum**) of S, if B has the following two properties:
(i) B is an upper bound of S.
(ii) No number less than B is an upper bound for S.

We denote supremum of S as $\sup S$.

Example 1: $S = (0, 4]$. Then $\sup S = 4$
Example 2: $S = (0, 4)$. Once again $\sup S = 4$

Supremum and Maximum

Observation 3: Supremum of a set is unique.

Observation 4: Supremum of S may not belong to S.

If $\sup S \in S$ then we call $\sup S$ as the maximum of S

ϵ-definition of Supremum

Take a non-empty set S.

A real number B is called a **least upper bound** (or **supremum**) of S, if B has the following two properties:
(i) B is an upper bound of S.
(ii) For every $\epsilon > 0$, there exists an element $a \in S$ such that $a > B - \epsilon$.

Equivalence of Two Definitions of Supremum

Theorem: Two definitions of supremum are equivalent.

Proof To show that the definitions are equivalent, we need to show the following:
(a) Take a set S and assume that B satisfies the original definition. We have to show that B satisfies the ϵ-definition.
(b) Take a set S and assume that B satisfies the ϵ-definition. We have to show that B satisfies the original definition.

Proof of (a): We prove by contradiction. Suppose B fails the ϵ-definition. That is, there exists $\epsilon^* > 0$ such that for all $a \in S$, $a \leq B - \epsilon^*$. That implies $B - \epsilon^*$ is an upper bound of S. However B is the supremum by the original definition. So a smaller number, namely $B - \epsilon^*$, can not be an upper bound of S.
We have reached a contradiction.

Proof of (b): Once again we prove by contradiction. Suppose B fails the original definition. That is, there is a $B' < B$ such that B' is also an upper bound of S. Hence $B' \geq x$ for all $x \in S$. Choose $\bar{\epsilon} > 0$ such that $B > B - \bar{\epsilon} > B'$. Therefore $B - \bar{\epsilon} > B' \geq x$ for all $x \in S$. Thus B fails the ϵ-definition at $\bar{\epsilon}$.
We have reached a contradiction.

Infimum

We can define lower bound and greatest lower bound (or infimum) in a similar fashion

A real number L is called a **greatest lower bound** (or **infimum**) of a non empty set S, if L has the following two properties:
(i) L is a lower bound for S, that is $L \leq x \, \forall x \in S$
(ii) No number greater than L is a lower bound for S.

We denote infimum of S as $\inf S$.

If $\inf S \in S$ then we call $\inf S$ the minimum of S

Completeness Axiom

Do all sets have supremum? Nope

Completeness Axiom

Every **non empty** set of real numbers which is **bounded above** has a supremum. Similarly, every **non empty** set of real numbers which is **bounded below** has an infimum.

Algebra of Supremum

- (Reflection) Let S be a non-empty set. Define $-S = \{-x \mid x \in S\}$. $B = \sup S \Leftrightarrow -B = \inf -S$.
- (Addition) Suppose $C = \{a + b \mid a \in A,\ b \in B\}$.
  Then $\sup C = \sup A + \sup B$.
- (multiplication) Suppose $\alpha > 0$ and $C = \{\alpha a \mid a \in A\}$.
  Then $\sup C = \alpha \sup A$.
- (Order) Suppose A and B are such that for every pair $(a, b) \in A \times B$, $a \leq b$, then $\sup A \leq \inf B$.